# The Anatomy of ICT Intangibles and the Rise in Markups

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# Overview

- **Rich administrative data**: Belgian firm-level B2B transactions linked to balance sheets w/ direct measures of ICT intangible investment

- **Empirical strategy**: stacked event study around lumpy spikes in ICT intangible investment

- **Main findings**: following ICT investment, large firms

  output ↑ employment ↑ fixed-cost share ↑ markups ↑

  while small firms respond little and net profit rates do not persistently rise

- **Interpretation**: higher markups reflect scalable technology and a shift from marginal toward fixed costs *rather than* greater market power

# My Discussion

1. Build a simple model to rationalize the main findings
    - Lower marginal costs
    - Adoption concentrated in large firms
    - Higher markups
    - Little change in profits

2. Use the model to interpret why markups rise following ICT investment

3. Consider the interaction between technology and market power

# Three Ingredients

1. **CES inputs**: intangibles lower the effective cost of input bundle
   (Brynjolfsson, Rock, and Syverson, 2021; De Ridder, 2024)

2. **Non-convex technology**: fixed costs make intangibles worthwhile only at scale
   (Buera, Kaboski, and Shin, 2011; Midrigan and Xu, 2014)

3. **Non-CES demand**: demand becomes less elastic as firms expand
   (Kimball, 1995; Dotsey and King, 2005; Klenow and Willis, 2016)

# Simple Model

- Firms differ in productivity $z$ and face downward-sloping demand $p(y)$
- Production uses a CES bundle of tangible and intangible capital
- Any $k^I > 0$ requires paying $F$

$$\pi = \max_{y,k^T,k^I \geq 0} \left\{ p(y)y - r_T k^T - r_I k^I - F\mathbf{1}\{k^I > 0\} \right\}$$

$$y = zk^\eta \qquad k = \begin{cases} k^T & \text{if } k^I = 0 \\ \left[ (k^T)^\rho + \omega (k^I)^\rho \right]^{1/\rho} & \text{if } k^I > 0 \end{cases}$$

# Marginal Costs

- If $k^I = 0$, then $k = k^T$ and

$$MC_0(y,z) = \frac{1}{\eta} r_T z^{-1/\eta} y^{1/\eta - 1}$$

- If $k^I > 0$, then $\frac{k^T}{k^I} = \left(\omega \frac{r_T}{r_I}\right)^{\frac{1}{\rho-1}}$ and

$$\begin{aligned} MC_1(y,z) &= \frac{1}{\eta}\left[r_T^{\frac{\rho}{\rho-1}} + \omega^{\frac{1}{1-\rho}} r_I^{\frac{\rho}{\rho-1}}\right]^{\frac{\rho-1}{\rho}} z^{-1/\eta} y^{1/\eta-1} \\ &= MC_0(y,z)\left[1 + \omega^{\frac{1}{1-\rho}}\left(\frac{r_I}{r_T}\right)^{\frac{\rho}{\rho-1}}\right]^{\frac{\rho-1}{\rho}} \end{aligned}$$

$$\boxed{k^I > 0 \quad \Rightarrow \quad MC \downarrow} \qquad (\text{for } \; 0 < \rho < 1)$$

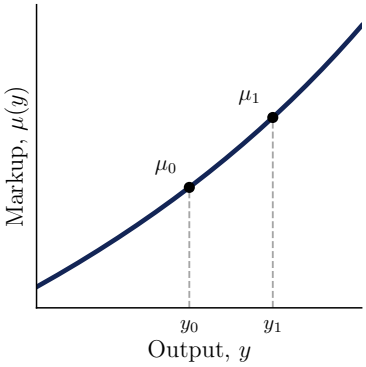
# Markups

WLOG: $p(y) = \alpha - \beta y$

$MR(y) = \alpha - 2\beta y$ $\qquad$ $MR(y) = MC(y)$

$$\mu(y) = \frac{p(y)}{MC(y)} = \frac{\alpha - \beta y}{\alpha - 2\beta y}$$

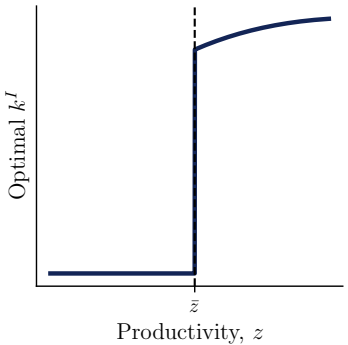
$$\mu'(y) = \frac{\alpha\beta}{(\alpha - 2\beta y)^2} > 0$$



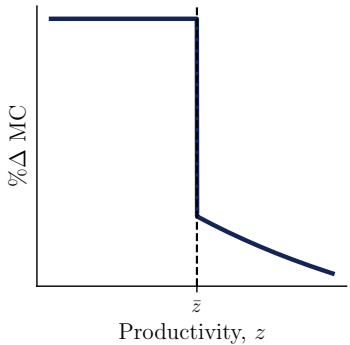

$$\boxed{k' > 0 \;\Rightarrow\; y \uparrow \Rightarrow\; \mu \uparrow}$$

# Predictions

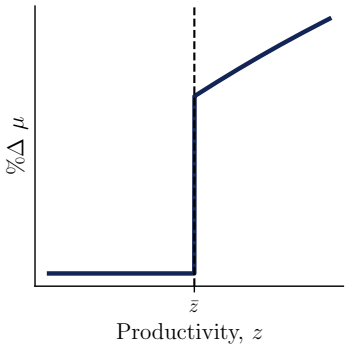
**Only sufficiently large firms adopt**




**Adoption lowers marginal cost**

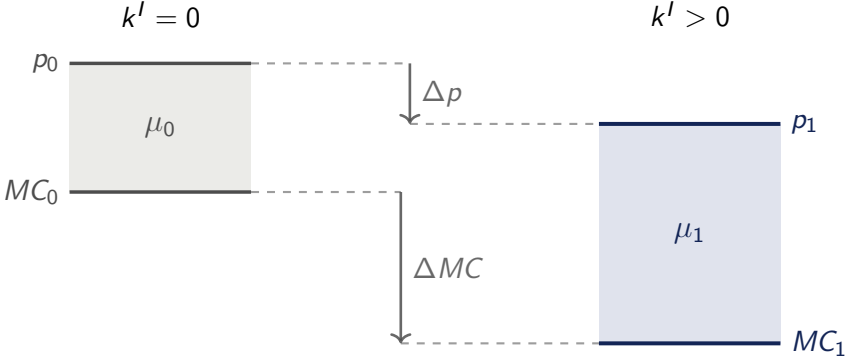



**Markups rise for adopters**



# Predictions

For $z > \bar{z}$:




$$y \uparrow \Rightarrow \; p \downarrow \quad \text{but} \quad |\Delta MC| > |\Delta p| \quad \text{so} \quad \mu \uparrow$$

Profits unchanged if $\boxed{\Delta\pi \equiv \pi_1 - \pi_0 = F}$

# Taking Stock

- Cost-side findings straightforward to rationalize: CES inputs lower marginal costs, fixed costs generate adoption at scale
- Higher markups require additional demand-side ingredient: as firms expand demand must become less elastic
- Implies greater market power among firms that benefit from the technology
- Technology adoption and market power can be complementary

# Conclusion

- Very nice paper!
- Unusually rich data and elegant empirical evidence
- Empirical findings fit nicely in a simple theoretical framework
- Theory can help sharpen the mechanism

# References


- Brynjolfsson, E., D. Rock, and C. Syverson (2021). "The Productivity J-Curve: How Intangibles Complement General Purpose Technologies." *American Economic Journal: Macroeconomics*, 13(1), 333–372.
- Buera, F. J., J. P. Kaboski, and Y. Shin (2011). "Finance and Development: A Tale of Two Sectors." *American Economic Review*, 101(5), 1964–2002.
- De Ridder, M. (2024). "Market Power and Innovation in the Intangible Economy." *American Economic Review*, 114(1), 199–251.
- Dotsey, M. and R. G. King (2005). "Implications of State-Dependent Pricing for Dynamic Macroeconomic Models." *Journal of Monetary Economics*, 52(1), 213–242.
- Kimball, M. S. (1995). "The Quantitative Analytics of the Basic Neomonetarist Model." *Journal of Money, Credit and Banking*, 27(4), 1241–1277.
- Klenow, P. J. and J. L. Willis (2016). "Real Rigidities and Nominal Price Changes." *Economica*, 83(331), 443–472.
- Midrigan, V. and D. Y. Xu (2014). "Finance and Misallocation: Evidence from Plant-Level Data." *American Economic Review*, 104(2), 422–458.